Qingyang Shu

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Research Interests: Machine Learning, Multi-view Learning, Data Mining, Applied Statistics

Educational Experiences

Southwestern University of Finance and Economics, Chengdu, P.R. China 2025.09 - Present

- *Master of Economics, Applied Statistics* | Score: 90.00/100
- Core Courses: Fundamentals of Data Science Statistics, Statistical Inference and Data Science Research, Machine Learning and Case Studies in Practice, etc.

Jiangxi University of Finance and Economics, Nanchang, P.R. China 2021.09 - 2025.06

- *Bachelor of Science, Applied Statistics* | Score: 90.03/100
- Core Courses: Mathematical Analysis, Advanced Algebra, Probability Theory, Mathematical Statistics, Statistical Machine Learning, Python Language and Data Analysis, etc.

Publications

[1] **Qingyang Shu**, Mingchang Cheng*, Jianmei Ren, Tiefeng Ma, and Aihua Han. 2026. *Step-wise incomplete multi-view clustering through joint utilization of structure and completion graphs*. Information Fusion. (**JCR Q1, IF:17.4**) DOI: https://doi.org/10.1016/j.inffus.2026.104268

[2] Tingjin Luo*, Mengyuan Tong, **Qingyang Shu**, Yueying Liu, Hao Zhou, and Zhen Wang. 2026. *Class: deep partial label feature selection with cluster-guided disambiguation and structured sparsity*. In Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.1 (KDD'26). (**Top-tier Conference, CCF-A**) DOI: https://doi.org/10.1145/3770854.3780163

Manuscripts

[1] **Qingyang Shu**, Jianmei Ren*, Mengyuan Tong, Mingchang Cheng, and Tiefeng Ma. 2026. *Multi-view positive–unlabeled learning with semi-supervised clustering and label disambiguation*. The 41st AAAI Conference on Artificial Intelligence. (**Top-tier Conference, CCF-A**) (Under Review)

[2] **Qingyang Shu**, Jianmei Ren, Mingchang Cheng*, and Tiefeng Ma. 2026. *Structure-guided incomplete multi-view clustering through consistency–complementarity coordination*. IEEE Transactions on Multimedia. (**JCR Q1, IF:9.9**) (Under Review)

[3] **Qingyang Shu**, Jianmei Ren, Aihua Han, Mingchang Cheng, Ling Zhou*, and Tiefeng Ma. 2026. *Can recidivism risk among parolees be reliably identified? — An empirical study using large language models and positive-unlabeled learning*. Journal of Interpersonal Violence. (**JCR Q1, IF:2.2**) (Under Review)

[4] Yao Dong, **Qingyang Shu***, and Mengyuan Tong. 2026. *A short-term probabilistic wind power forecasting integrating deep reinforcement learning and multi-objective bayesian optimization*. International Journal of Electrical Power and Energy Systems. (**JCR Q1, IF:5.3**) (Under Review)

[5] Mengyuan Tong, **Qingyang Shu**, and Tingjin Luo*. 2026. *DCR: dual-correction confidence refinement for coupled instance-dependent long-tailed partial-label learning*. The 41st AAAI Conference on Artificial Intelligence. (**Top-tier Conference, CCF-A**) (Under Review)

[6] Jianmei Ren, **Qingyang Shu**, Mingchang Cheng, and Tiefeng Ma*. 2026. *A hierarchical multi-scale framework for interpretable signal segmentation*. Computers & Industrial Engineering. (**JCR Q1, IF:7.3**) (Under Review)

[7] Jianmei Ren, Aihua Han*, **Qingyang Shu**, Tiefeng Ma, and Shuangzhe Liu. 2026. *Quantifying causal responsibility under unobserved confounding via a key-segment-based method*. IEEE Transactions on Neural Networks and Learning Systems. (**JCR Q1, IF:9.7**) (Under Review)

Research Projects

Multi-view Unsupervised and Semi-supervised Learning 2025.09 - Present

- Proposed several algorithms for incomplete multi-view clustering based on graph representation learning, addressing the problem of missing-view completion while effectively exploiting the consistency and complementarity among multiple views.

- Extended positive-unlabeled learning to multi-view scenarios and proposed a framework that integrates clustering and label disambiguation to address the challenges of unreliable negative sample selection and strong dependence on class priors in existing methods.
- Integrated large language models with multi-view positive-unlabeled learning to develop a criminal justice risk assessment framework that constructs multidimensional risk representations and identifies latent recidivism risk under incomplete observations.

Estimation of the Seabed Topography of the Western Pacific Ocean 2024.09 – 2025.06

- Utilized oceanographic observations and satellite altimetry data to compute gravity anomalies using the inverse Stokes formula and inverse Vening-Meinesz formula.
- Provided high-quality gravity anomaly data to support subsequent seabed topography inversion.

Deep Learning based Wind Power Interval Forecasting 2023.09 – 2025.06

- Developed a wind power interval forecasting system using TCN-Transformer and BiLSTM models for temporal feature modeling.
- Integrated predictions via reinforcement learning and performed multi-objective Bayesian optimization for automated hyperparameter tuning, improving the reliability and stability of prediction intervals.

Awards & Scholarships

- Outstanding Graduate Student, Southwestern University of Finance and Economics 2026.03
- Outstanding Graduate from Jiangxi University of Finance and Economics 2025.06
- Lixin Scholarship (Top 1%) 2024.12
- First Prize, National Undergraduate Statistical Modeling Competition 2024.08
- First Prize, National Undergraduate Market Research and Analysis Competition 2024.06
- Dongyi Scholarship (Top 1%) 2023.12
- Third Prize, MathorCup Mathematical Modeling Competition 2023.06
- H Award, HuaShu Cup Mathematical Modeling Competition 2023.02

Skills

- Language: English (IELTS: 7; CET-6: 554), Chinese (Native)
- Programming: Python, R, Matlab.
- Tools: Latex, MS Office, SPSS, etc.